

Exploratory Data Analysis

Group029 | Standardised historical retail dataset

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Future ML questions

1. Problem context, scope and data-preparation assurance

Scope. Two historical JSON/XML exports describe customers, products, orders, order items, deliveries and customer reviews. The analysis uses the six submitted standardised CSVs and preserves their canonical grains. Model training is outside A1.

Four material transformation decisions

Decision	Implementation
Relational grain	Keep six tables at their prescribed grains; do not flatten one-to-many relationships into duplicated order rows.
Normalise before reconcile	Preserve ID case/leading zeros; standardise dates, timestamps, booleans, currency and percentages before cross-source comparison.
Published arithmetic	Round line revenue, aggregate order price, report included GST separately, then apply discount and delivery charges in the specified order.
Bounded text cleaning	Parse JSON/XML structurally before regex; preserve multilingual clean text and derive a separate Latin-analysis field.

Material validation evidence

Validation ID	Evidence	Observed
VAL-FLOW-ORD-01	Source order-key union reconciled to canonical output	PASS
VAL-FLOW-ORD-02	Normalised cross-source order conflicts	0; PASS
VAL-ARITH-TOTAL-01	Discount/delivery order-total reconciliation within AUD 0.01	PASS
VAL-PK-REV-01	Review primary-key completeness and uniqueness	7,000 rows; PASS
VAL-FK-REV-01	Review relationships to orders/items/products/customers	0 orphan keys; PASS
VAL-TEXT-REV-01	Published and additional multilingual/reference text cases	PASS

The complete lineage remains in the 111-row mapping CSV; the complete executable validation register remains in the solution notebook. The EDA notebook independently rechecks all six primary keys and eight required foreign-key relationships before analysis.

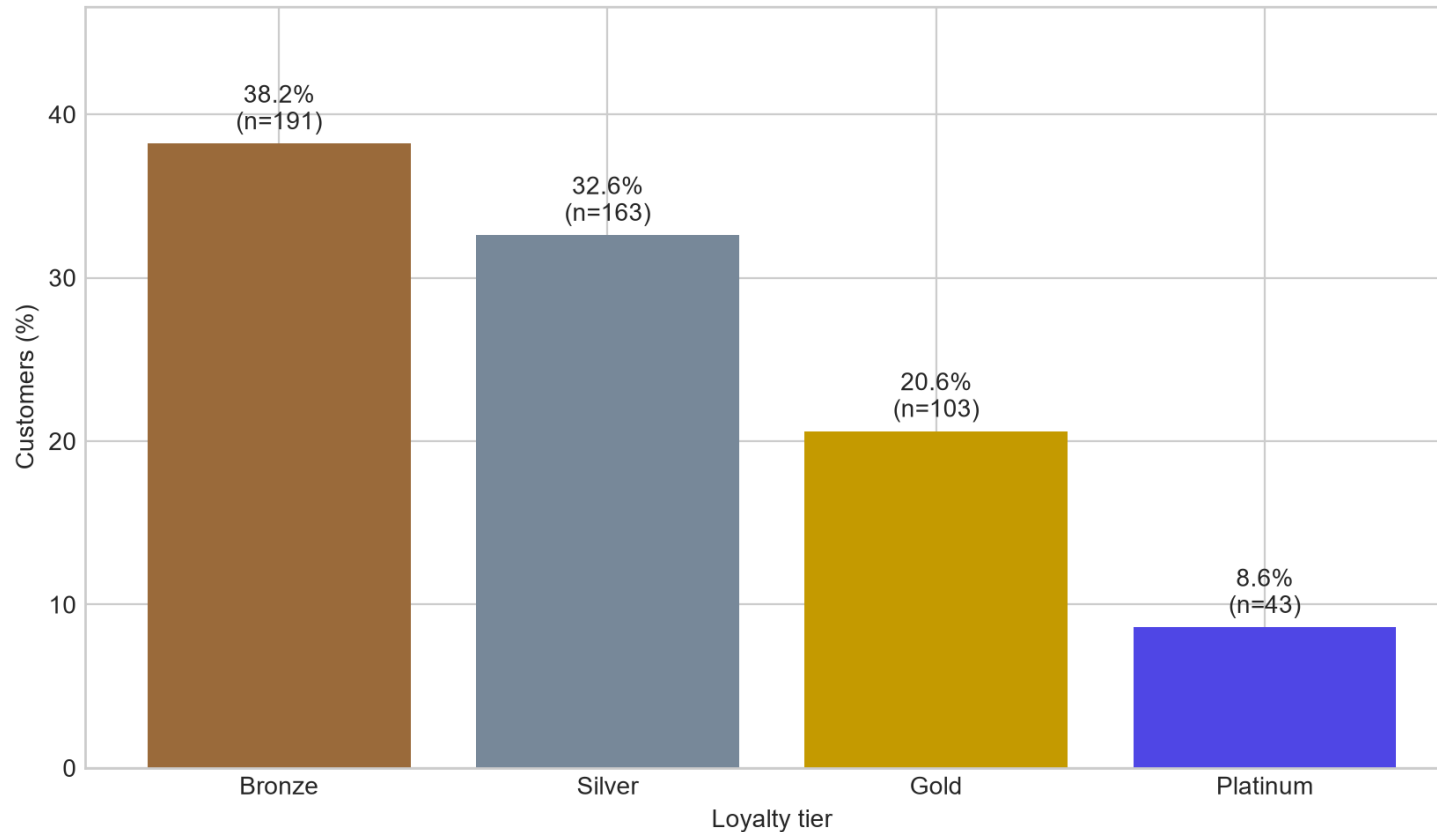
2. Assessed EDA - Figure 1: Customer composition

Question: What is the composition of the customer base by loyalty tier?

Unit/denominator: One canonical customer; 500 customers. Tables: customers; no join.

Limitation: Tier size does not itself measure value or engagement, and the tier rules are not supplied.

Figure 1. Customer composition by loyalty tier



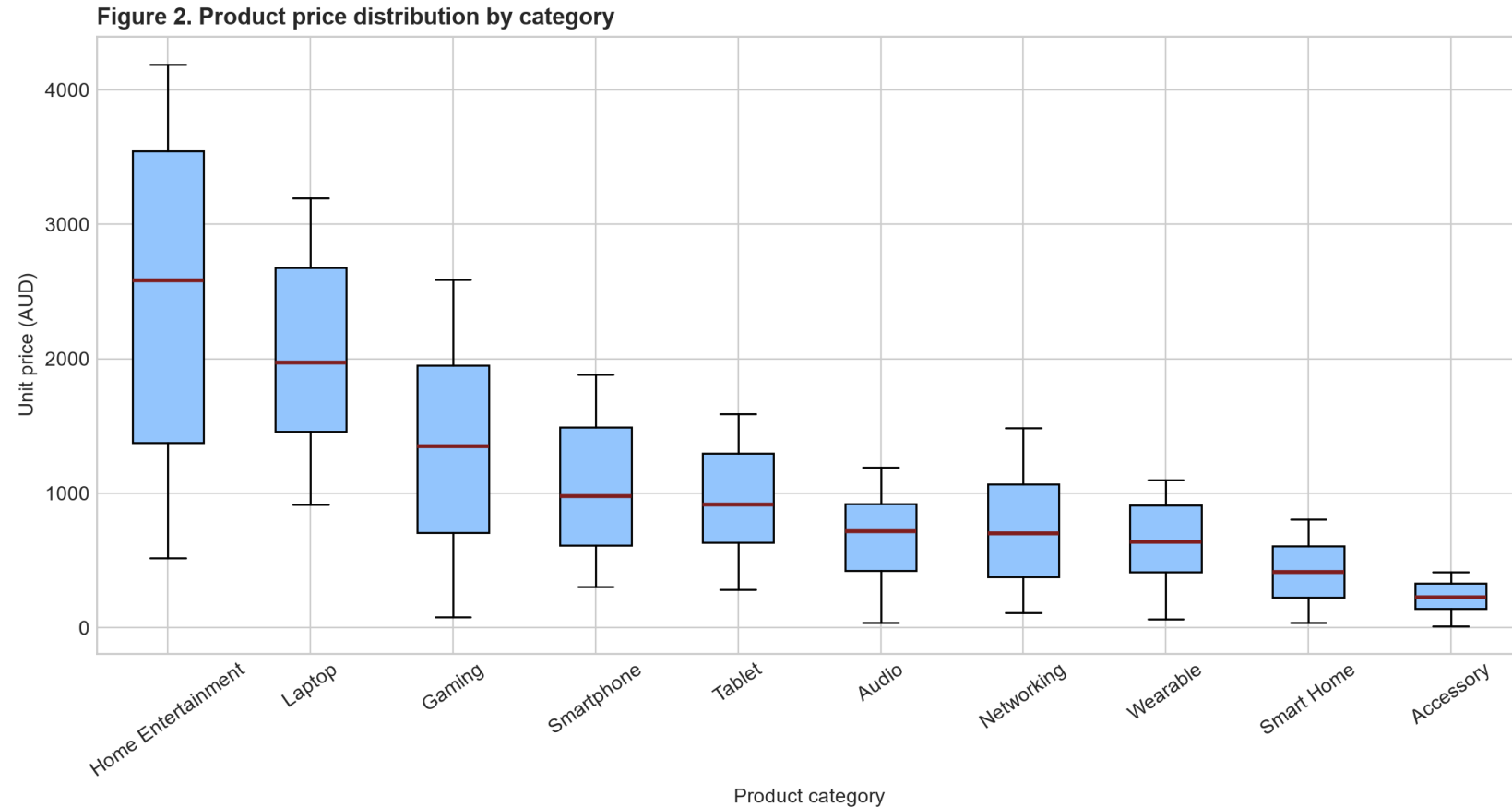
Finding 1. Bronze is the largest tier with 191 of 500 customers (38.2%), followed by Silver with 163 (32.6%); Platinum contains 43 (8.6%). This matters for targeting and capacity estimates, but tier size does not measure profitability. Compare tier rules and customer outcomes before using tier membership operationally.

2. Assessed EDA - Figure 2: Product price distribution

Question: How does listed product price vary by category?

Unit/denominator: One canonical product; 1,000 products, 100 per category. Tables: products; no join.

Limitation: Listed prices are not transaction-weighted realised prices; chart outliers are hidden only for readability.

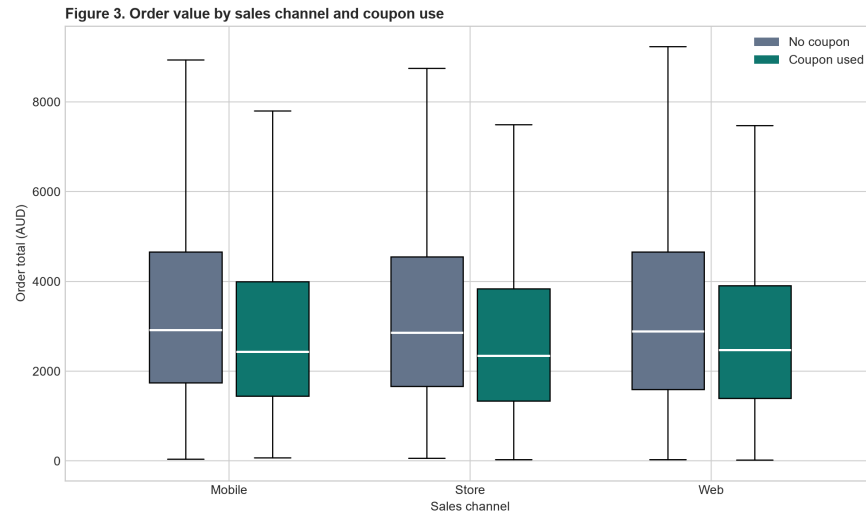


Finding 2. Each category contains 100 products, yet median listed price ranges from AUD 227.97 for Accessory to AUD 2,585.08 for Home Entertainment. This signals distinct catalogue positioning. Stock availability, discounting and realised product mix must be added before revenue forecasting.

2. Assessed EDA - Orders, coupons and time

Figure 3. Order value by channel and coupon use

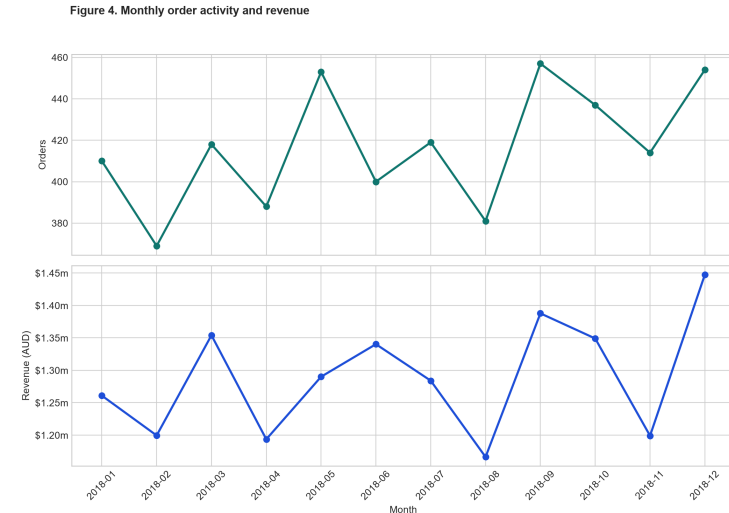
Question: How do order values differ by sales channel and coupon use? Unit/denominator: One order; n=5,000. Tables: orders only; no join. Limitation: Post-discount totals and non-random coupon allocation prevent causal interpretation.



Finding 3. Coupon-order medians are about AUD 411-519 below the matching no-coupon medians across Mobile, Store and Web. The coupon groups contain 1,020, 1,103 and 1,102 orders respectively. The mechanical discount and non-random allocation mean this is not evidence that coupons reduce purchasing; incrementality needs a controlled or matched design.

Figure 4. Monthly order activity and revenue

Question: How do monthly order counts and revenue vary? Unit/denominator: Calendar-month aggregates from 5,000 orders. Tables: orders only; no join. Limitation: One year and unequal month lengths cannot establish stable seasonality.



Finding 4. September has the most orders (457), while December has the highest revenue (AUD 1,447,431.47 from 454 orders). Revenue therefore reflects order value and mix as well as volume. Inspect item/category composition before inventory or staffing changes.

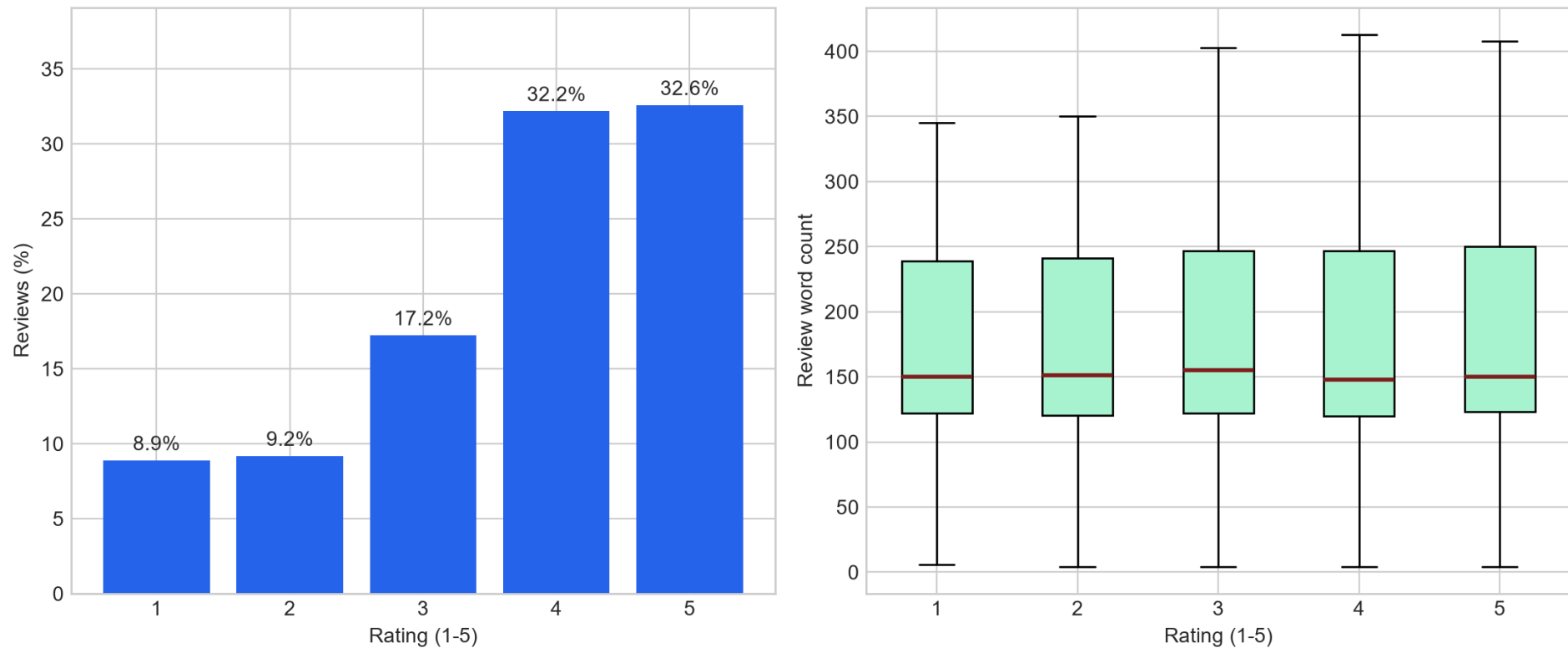
2. Assessed EDA - Figure 5: Review rating and length

Question: How are ratings distributed, and does review length differ by rating?

Unit/denominator: One canonical review; 7,000 reviews. Tables: product_reviews; no join.

Limitation: Non-reviewers are absent and review length is not a direct measure of review quality.

Figure 5. Review rating composition and length



Finding 5. Ratings 4-5 account for 4,532 of 7,000 reviews (64.7%), while ratings 1-2 account for 1,263 (18.0%). Median length stays between 148 and 155 words across ratings, so word count alone offers little separation. Selection and solicitation effects may shape the observed distribution.

2. Assessed EDA - Figure 6: Category demand and review outcomes

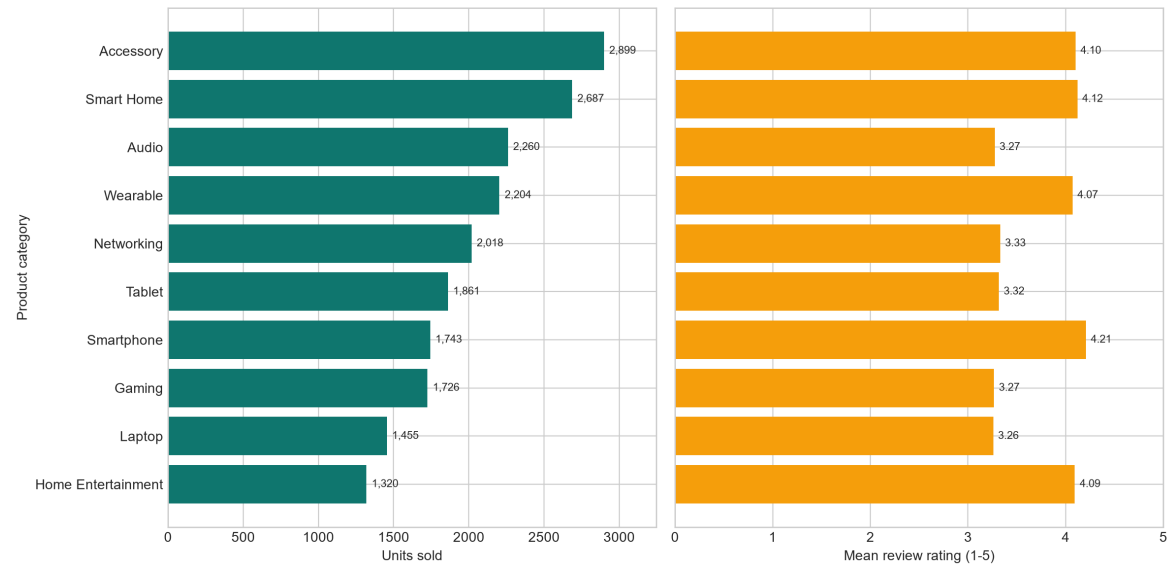
Question: Which categories combine stronger unit demand with stronger or weaker review outcomes?

Unit/denominator: 15,706 order items and 7,000 reviews, separately aggregated to category.

Tables/joins: order_items.product_id and product_reviews.product_id to products.product_id; both many-to-one joins retain their original row counts.

Limitation: Availability, exposure, price and review-selection effects are not controlled.

Figure 6. Product-category demand and review outcomes



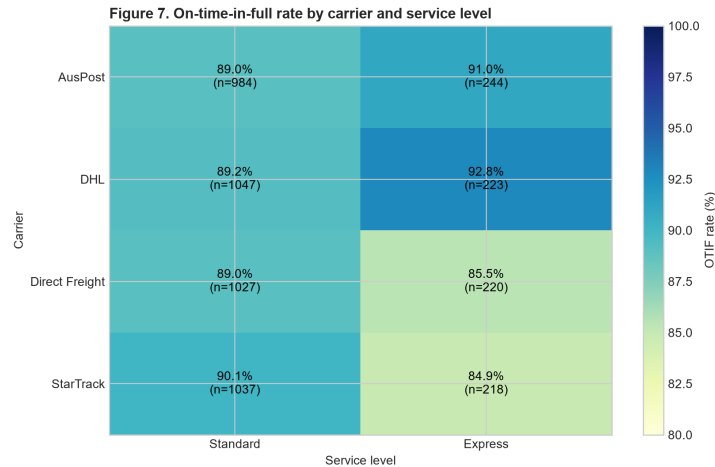
Finding 6. Category mean rating ranges from 3.26 for Laptop (678 reviews) to 4.21 for Smartphone (694). Similar review counts reduce simple sample-size concerns, but brand, price, product age and purchaser mix may explain the gap. Inspect within-category and product-level variation before attributing the result to category.

Finding 7. Accessory has the highest unit volume (2,899) and a 4.10 mean rating, while Home Entertainment has 1,320 units and a similar 4.09 rating. Volume and average rating do not form a simple trade-off. The two fact tables were aggregated separately, avoiding order-item × review multiplication.

2. Assessed EDA - Delivery operations

Figure 7. OTIF by carrier and service level

Question: How does OTIF vary by carrier and service level? Unit/denominator: One delivery; n=5,000. Tables: deliveries only; no join. Limitation: Route/destination mix may confound carrier comparisons.

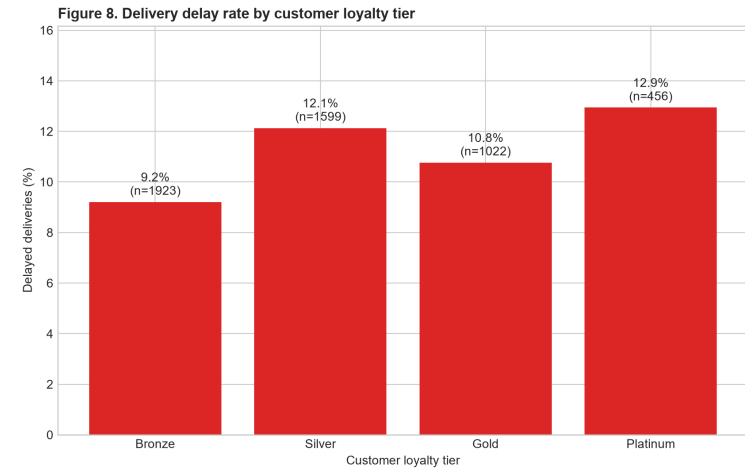


Finding 8. Among 5,000 deliveries, 4,460 (89.2%) are on time in full and 540 (10.8%) are delayed. Delays remain large enough for exception monitoring, but controllable carrier causes are not isolated.

Finding 9. DHL Express records 92.8% OTIF (n=223) versus 84.9% for StarTrack Express (n=218), a 7.9-point gap. Standard services are closer at about 89.0%-90.1%. Adjust for route mix before changing allocation.

Figure 8. Delay rate by loyalty tier

Question: Does delivery delay rate differ by loyalty tier? Unit/denominator: One matched order/delivery; n=5,000. Tables/joins: orders to customers and orders to deliveries; checked sequential joins retain all order IDs. Limitation: Tier is observational and is not a causal explanation of delay.



Finding 10. Bronze records 177 delays among 1,923 orders (9.2%), versus 59 among 456 Platinum orders (12.9%). Tier is not causal and must not be used to deprioritise customers. Test operational variables first and monitor fairness.

3. Five future machine-learning questions (1-2)

ML Question 1 - Forecast weekly order volume by sales channel

Element	Response
EDA evidence	Figures 3-4 show three active channels and month-to-month order variation.
Decision / unit	Allocate short-term warehouse labour; one week × sales channel.
Type / target	Forecasting; next-week order count by channel.
Predictors	Lagged weekly counts, rolling summaries and calendar/channel fields available before cutoff.
Validation / metric	Expanding-window temporal validation; MAE and WAPE against a naive baseline.
Risks	Future observations in rolling features are leakage; one year, missing campaigns/stock and drift limit deployment.

ML Question 2 - Identify low-rating review risk after fulfilment

Element	Response
EDA evidence	Figures 5-6 show an 18.0% low-rating share and category rating differences.
Decision / unit	Prioritise post-purchase service; one fulfilled order item.
Type / target	Binary classification; eventual rating of 1-2.
Predictors	Product/order/delivery/customer-history variables available by fulfilment.
Validation / metric	Time-based split; PR-AUC, low-rating recall and calibration.
Risks	Review text/rating-derived fields are leakage; non-reviewers create selection bias; audit subgroup fairness.

3. Five future machine-learning questions (3-5)

ML Question 3 - Estimate future product demand

Element	Response
Evidence	Figures 2 and 6: category price and unit-volume differences.
Decision / unit	Replenishment; product × future period.
Type / target	Regression/forecasting; future units sold.
Predictors	Category, price, cost, launch age, warranty, prior demand and season, frozen before cutoff.
Validation	Rolling temporal split; MAE/WAPE.
Risks	Future reviews/promotions are leakage; stock-outs censor demand.

ML Question 4 - Discover customer segments

Element	Response
Evidence	Figures 1 and 8: uneven tiers and descriptive operational differences.
Decision / unit	Service research; one customer.
Type / objective	Clustering using pre-period value, frequency and preference attributes.
Validation	Earlier/later cohort stability; silhouette and cluster-size checks.
Risks	Proxy attributes may create unfair segments; require human review and minimum sizes.

ML Question 5 - Predict late delivery at dispatch

Element	Response
Evidence	Figures 7-8: delay prevalence and segment variation.
Decision / unit	Proactive intervention; one dispatch.
Type / target	Binary classification; delay_days > 0.
Predictors	Carrier, service, promised days, distance, signature, expedited flag, warehouse, total and season.
Validation	Time split; PR-AUC, recall and calibration.
Risks	Delivered date/status, delay reason and tracking outcomes leak; audit fairness and drift.

4. Limitations and conclusion

Limitations

- The analysis covers one historical year, so monthly movement cannot establish recurring seasonality or future behaviour.
- All relationships are observational. Coupon use, carrier, service level, customer tier and product category were not randomly assigned.
- Reviews represent only customers who submitted reviews; availability, exposure, stock-outs, marketing campaigns and acquisition costs are not observed.
- Category and subgroup summaries may hide within-group variation. Small Platinum and carrier-service groups create greater uncertainty.
- Join multiplication was checked for Figures 6 and 8, but operational or model decisions still require fresh-period validation and monitoring.

Conclusion

Group029's standardised relational outputs support a coherent view of customers, products, transactions, reviews and delivery operations. The strongest descriptive signals are large category price differences, ratings concentrated at the high end, monthly variation in activity and revenue, and non-trivial delivery-delay differences across operational segments. These results identify feasible forecasting, classification, regression and clustering questions, but they do not establish causation. Any later model or business change should use time-aware validation, leakage controls, explicit join-grain checks and fairness monitoring.

References

1. Monash University. (2026). FIT5196 S2 2026 Assessment 1 Specification.
2. Monash University. (2026). FIT5196 S2 2026 Assessment 1 Marking Rubric.
3. The pandas development team. (2026). pandas documentation. <https://pandas.pydata.org/docs/>
4. Matplotlib Development Team. (2026). Matplotlib documentation. <https://matplotlib.org/stable/>

Page-limit note. Pages 2-11 contain the assessed material from introduction through conclusion (10 pages). The cover page and this references page are excluded from the limit under the assignment specification.